

Words to Embeddings

Sunday, September 14, 2025 11:41 AM

Project Gutenberg, easy data source

Prediction based methods:

Continuous Bag of Words

Skip grams

Looks at k words around a center word

Weights are initially set randomly

Softmax

Natural log function

Aren't true probabilities, but can be treated as such

Continuous function

Argmax

Discontinuous

Functions need to be continuous so we can back propagate over them

We want the weights going from layer 1 to layer 2, this is how we get our embeddings

Skip Gram example:

And = [0,0,1,0,0]

Pink flowers and blue diamonds

-> [0.25,0.25,0,0.25,0.25]

pink flowers and pink diamonds

-> [0.375, 0.25, 0, 0.125, 0.25]

Back propagation helps push your weights towards the local maximum

Cont. Bag of Words example

Pink = 0.25, flowers = 0.25, and = 0, blue = 0.25, diamonds = 0.25

-> trying to predict the word in the middle

Pink = 0, flowers = 0, and = 1, blue = 0, diamonds = 0

Count based models

Count up all words in a document and put into a vocab vector

Word counts can be used to group documents